1. MAIZE PRODUCTION

Areas Suitable for Maize Production

Maize is one of the world's most important cereal crops and grows best in warm climates with fertile, well-drained soils.

In Kenya

The major maize-producing regions include:

- **Trans Nzoia County** (largest producer)
- **Uasin Gishu County**
- **Nakuru County**
- **Nandi County**
- **Bungoma County**
- **Kakamega County**
- **Narok County**
- **Bomet County**
- **Elgeyo Marakwet County**
- Parts of **Meru**, **Embu**, and **Laikipia** counties

Globally

Major maize-producing countries include:

- United States
- China
- Brazil
- Argentina
- India
- Mexico
- South Africa

. Climate Requirements for Maize Production

	Requirement
Temperature	18–30°C (optimum 25–30°C)
Rainfall	500–1,200 mm well distributed during the growing season

	Requirement
Altitude	Sea level to over 2,500 m depending on the variety
Sunlight	Full sunlight (6–8 hours daily)
Soil	Deep, fertile, well-drained loam soils with pH 5.5–7.5

Effects of unsuitable climate

- Low rainfall causes wilting and poor grain filling.
- Excess rainfall causes waterlogging and root diseases.
- Frost damages young plants.
- Very high temperatures reduce pollination and grain formation.

Management Practices in Maize Production

Before Planting

- Select suitable land.
- Prepare land early through ploughing and harrowing.
- Choose certified, high-yielding seed varieties.
- Conduct soil testing where possible.
- Apply farmyard manure or basal fertilizer (DAP or NPK).

CASE

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Purpose:

- Loosens the soil for root penetration.
- Improves water infiltration.
- Controls weeds before planting.
- Creates a fine seedbed for good germination.
-

During Planting

- Plant at the onset of rains.
- Use proper spacing:
    - 75 cm × 30 cm (one plant per hole)
    - or 75 cm × 60 cm (two plants per hole)
- Plant seeds 3–5 cm deep

75 CM × 25 CM
SPACING
= HIGH YIELD!

Recommended practices

- Plant at the onset of rains.
- Use certified seeds.
- Maintain spacing of **75 cm × 30 cm**.
- Plant seeds **3–5 cm deep**.

Crop Management

- Thin seedlings to maintain the required plant population.
- Carry out timely weeding (2–3 times).
- Top dress with CAN or Urea when plants are knee-high.
- Earth up soil around plants for support.
- Monitor pests and diseases regularly.
- Practice crop rotation.
- Irrigate where rainfall is insufficient.

c) Fertilizer Application

Importance

- Reduces competition for nutrients, water, and sunlight.
- Minimizes pest and disease habitats.
- Improves crop gro

Irrigation

Purpose

- Ensures adequate moisture during dry periods.
- Improves grain filling.
- Prevents moisture stress.

Harvesting and Storage

Management

- Harvest when cobs are fully mature.
- Dry grain to about **13–14% moisture**.
- Store in clean, dry, airtight stores or hermetic bags.

Common Pests of Maize

a) Fall Armyworm

Symptoms

- Large holes in leaves.
- Ragged leaves.
- Frass (sawdust-like droppings) inside the whorl.
- Damaged growing point.

Control

- Scout fields regularly.
- Apply recommended insecticides.
- Encourage natural enemies.
- Destroy heavily infested plants.

b) Stem Borer

Symptoms

- Dead heart in young plants.
- Holes in stems.
- Weak stems that lodge easily.

Control

- Early planting.

- Crop rotation.
- Destroy crop residues.
- Apply recommended insecticides.

c) Cutworms

Symptoms

- Seedlings cut at ground level.
- Missing plants.

Control

- Early weeding.
- Field sanitation.
- Recommended insecticides.

d) Maize Weevil (Storage Pest)

Symptoms

- Holes in stored grains.
- Powdery grain.
- Weight loss.

Control

- Dry grain properly.
- Use hermetic storage bags.
- Store in clean, dry stores.
- Fumigate when necessary.

Pests and Their Control

Pest	Damage	Control/Remedy
Fall armyworm	Feeds on leaves and whorls	Scout fields regularly, handpick larvae where possible, apply recommended insecticides, encourage natural enemies, rotate crops.
Stem borers	Bore into stems causing dead hearts	Destroy crop residues, plant early, rotate crops, use resistant varieties, apply recommended insecticides.
Cutworms	Cut seedlings at ground level	Deep ploughing, field sanitation, hand collection, apply suitable insecticides.
Maize weevils (storage)	Damage stored grain	Dry grain properly, use airtight storage, treat grain with approved storage pesticides, use hermetic bags.

Common Diseases of Maize

a) Maize Streak Disease

Symptoms

- Yellow streaks running along leaves.
- Stunted growth.
- Poor cob development.

Control

- Plant resistant varieties.
- Control leafhopper vectors.
- Early planting.

b) Northern Leaf Blight

Symptoms

- Long grey-green lesions.
- Premature drying of leaves.

Control

- Crop rotation.
- Resistant varieties.
- Fungicides if severe.

c) Grey Leaf Spot

Symptoms

- Rectangular grey lesions.
- Reduced photosynthesis.
- Yield reduction.

Control

- Crop rotation.
- Destroy infected residues.
- Resistant varieties.

d) Common Rust

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Symptoms

- Brown or reddish pustules.
- Yellowing and drying of leaves.

Control

- Resistant varieties.
- Fungicides where necessary.

e) Maize Smut

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Symptoms

- Large swollen galls on ears, tassels, or stems.
- Black powdery spores.

Control

- Remove infected plants.
- Crop rotation.
- Resistant varieties.

f) Ear Rot

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Symptoms

- Mouldy kernels.
- Rotten cobs.
- Poor grain quality.

Control

- Harvest on time.
- Dry grain thorough

Pests and Their Control

Pest	Damage	Control/Remedy
Fall armyworm	Feeds on leaves and whorls	Scout fields regularly, handpick larvae where possible, apply recommended insecticides, encourage natural enemies, rotate crops.
Stem borers	Bore into stems causing dead hearts	Destroy crop residues, plant early, rotate crops, use resistant varieties, apply recommended insecticides.
Cutworms	Cut seedlings at ground level	Deep ploughing, field sanitation, hand collection, apply suitable insecticides.
Maize weevils (storage)	Damage stored grain	Dry grain properly, use airtight storage, treat grain with approved storage pesticides, use hermetic bags.

TEA PRODUCTION

Areas Suitable for Tea Production

Tea is mainly grown in cool, highland areas with reliable rainfall.

In Kenya

Major tea-producing areas include:

Kericho

Bomet

Nandi

Nyamira

Kisii

Kiambu

Murang'a

Nyeri

Kirinyaga

Meru

Embu

Tharaka Nithi

Other Leading Tea-Producing Countrie ;China,India

Sri Lanka

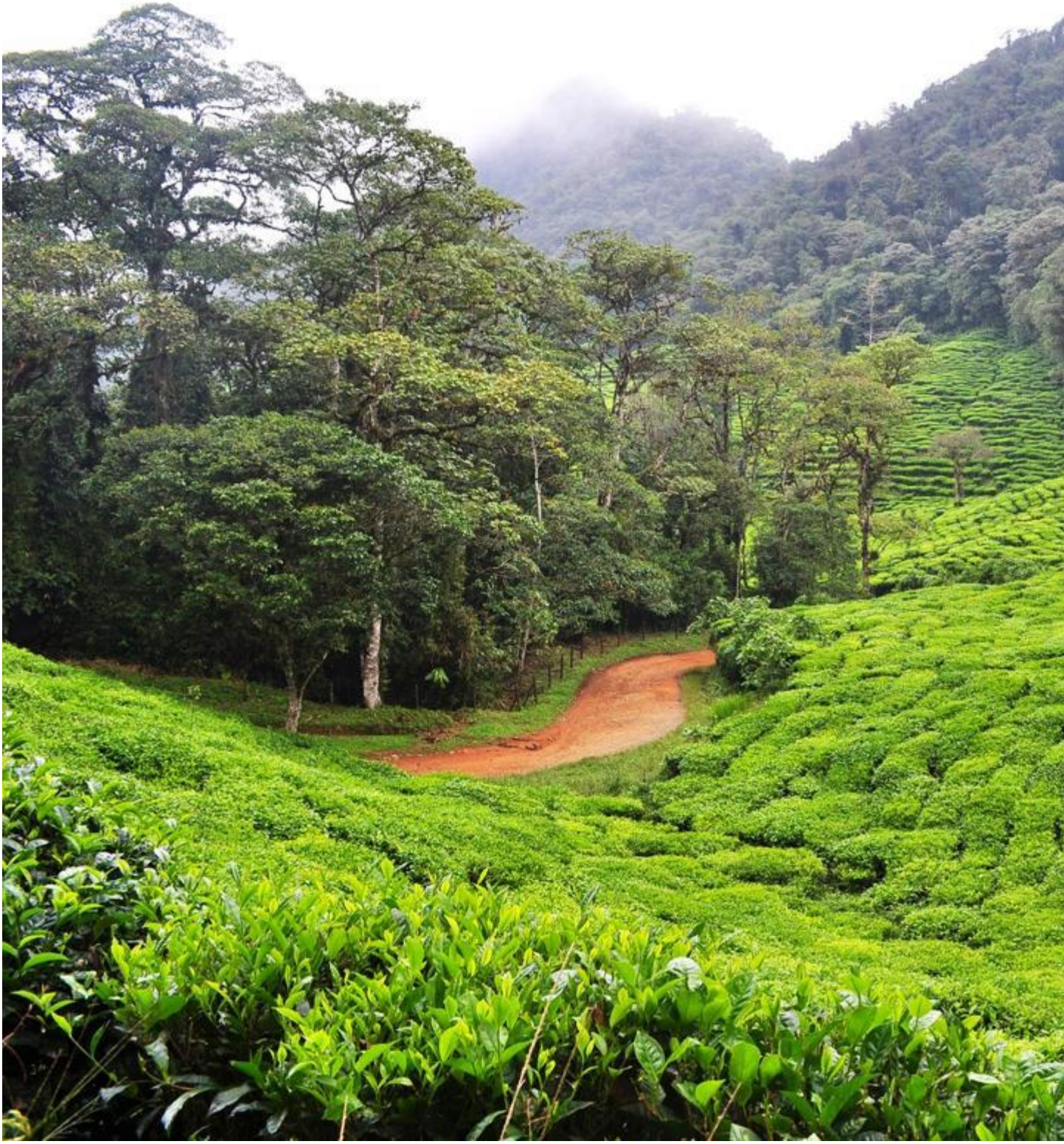
Uganda

Rwanda

Tanzania

- Vietnam

Climate Suitable for Tea Production



Tea grows best under the following climatic conditions:

- **Temperature:** 18–30°C (optimum is 20–25°C).
- **Rainfall:** 1,200–2,500 mm per year, well distributed throughout the year.
- **Altitude:** 1,500–2,700 metres above sea level.
- **Humidity:** High humidity encourages vigorous growth.

- **Sunshine:** Moderate sunshine with occasional cloud cover.
- **Soil:** Deep, fertile, well-drained acidic soils with a pH of **4.**

- Tea Production Practices

a) Land Preparation

Clear vegetation.

Remove tree stumps and stones.

Prepare planting holes.

Add manure or compost.

b) Planting

Plant healthy seedlings at the onset of rains.

Recommended spacing:

1.2 m × 0.75 m

(Spacing may vary with variety.)

c) Mulching

Conserves soil moisture.

Reduces weed growth.

Improves soil fertility.

d) Fertilizer Application

Apply compost or well-rotted manure.

Use nitrogen-rich fertilizers where recommended.

e) Weeding

Weed regularly to reduce competition.

f) Pruning

Remove old and unproductive branches.

Maintains a convenient plucking table.

Encourages new shoots.

g) Pest and Disease Control

Scout regularly.

Maintain field sanitation.

Use resistant varieties where available.

Apply recommended pesticides or fungicides only when necessary.

Harvesting

Pick the **top two leaves and a bud**.

Harvest every **7–14 days**, depending on growth conditions.

4. Common Tea Diseases

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Disease	Symptoms	Control
Blister blight	White blisters on young leaves	Use resistant varieties, prune infected shoots, apply fungicides where recommended
Armillaria root rot	Yellowing, wilting, root decay	Remove infected plants, improve drainage, avoid planting in infected sites
Grey blight	Brown or grey leaf spots	Remove infected leaves and apply recommended fungicides
Branch canker	Drying and dieback of branches	Prune affected branches and maintain field hygiene

Common Tea Pests

Pest	Damage	Control
Tea mosquito bug	Sucks sap, causing leaf damage and shoot dieback	Use recommended insecticides and prune affected shoots
Red spider mites	Cause yellowing, bronzing and drying of leaves	Encourage natural predators, avoid dusty conditions, apply acaricides if needed
Aphids	Suck plant sap and weaken plants	Introduce natural enemies or apply suitable insecticides
Scale insects	Reduce plant vigour by sucking sap	Prune heavily infested branches and use recommended insecticides

Rice Production

1. Introduction

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Rice (*Oryza sativa*) is one of the world's most important cereal crops and is the staple food for more than half of the global population. It is mainly grown in tropical and subtropical regions but can also be cultivated in temperate climates with adequate water and suitable temperatures.

Areas Where Rice is Produced

Worldwide

Major rice-producing countries include:

- China
- India
- Indonesia
- Bangladesh
- Vietnam
- Thailand
- Myanmar
- Philippines
- Pakistan
- Brazil

In Kenya

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Major rice-growing regions include:

- Mwea Irrigation Scheme (Kirinyaga County)
- Ahero Irrigation Scheme (Kisumu County)
- Bunyala Irrigation Scheme (Busia County)
- West Kano Irrigation Scheme
- Tana Delta
- Kwale County

- Lower parts of Meru and Embu (small-scale production)

Climatic Requirements

Factor	Requirement
Temperature	20–35°C (Optimum: 25–30°C)
Rainfall	1,000–2,000 mm annually or adequate irrigation
Altitude	Sea level to about 2,000 m
Sunshine	6–8 hours daily
Relative Humidity	60–80%
Water	Continuous water supply during vegetative growth

-

Types of Rice Farming

A. Irrigated Rice

- Grown under controlled irrigation.
- Highest yields.

B. Rain-fed Lowland Rice

- Depends on rainfall.
- Common in flood plains.

C. Upland Rice

- Grown without flooding.
- Requires well-drained soils.

D. Deep-water Rice

- Grown where water levels rise significantly during the season.

Land Preparation

Activities include:

- Clearing vegetation.
- Ploughing.
- Harrowing.
- Levelling.
- Puddling (mixing soil with water in irrigated systems).
- Constructing bunds to retain water.

Rice Varieties

Common varieties include:

- Basmati
- IR64
- NERICA
- Komboka
- BW196
- Sindano
- Hybrid rice varieties

Seed Selection

Good seed should be:

- Certified
- Disease-free
- High germination percentage
- Pure variety
- Free from weed seeds

Seed rate:

- 30–60 kg/ha (transplanted)
- 80–120 kg/ha (direct seeding)

Planting Methods

A. Direct Seeding

Seeds are planted directly in the field.

Advantages:

- Saves labour
- Faster planting
- Lower production cost

B. Transplanting

Seedlings are raised in nurseries for 21–30 days before transplanting.

Advantages:

- Better spacing

- Easier weed control
- Higher yields

Recommended spacing:

- 20 cm × 20 cm
- 25 cm × 25 cm

Water Management

Maintain:

- 2–5 cm water during early growth.
- 5–10 cm during vegetative stage.
- Drain fields before harvesting.

. Fertilizer Application

Common fertilizers:

Basal application:

- DAP
- NPK

Top dressing:

- Urea
- CAN (where appropriate)

Nutrients required:

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Zinc (Zn)

Weed Control

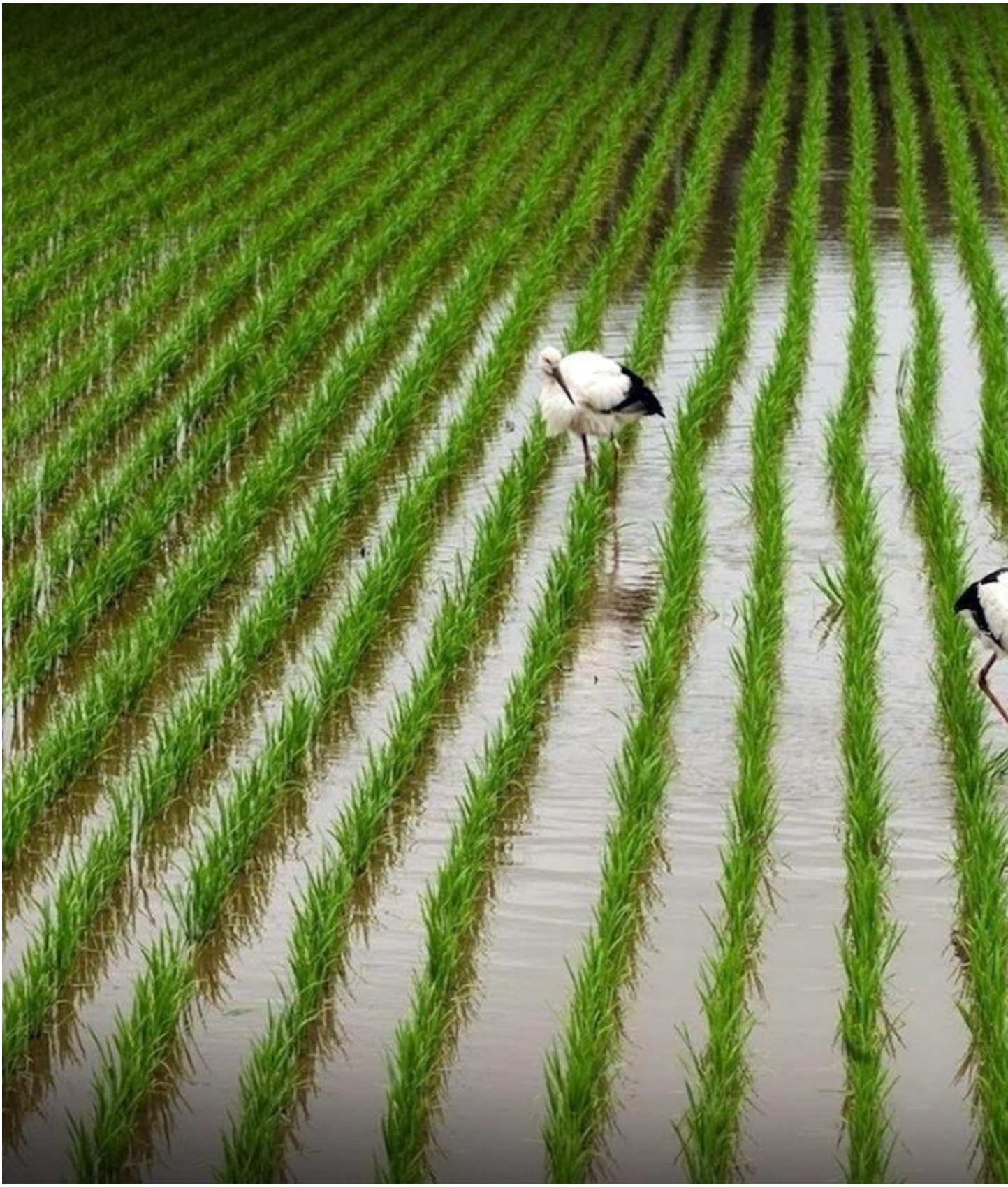
Methods include:

- Hand weeding
- Mechanical weeding
- Flooding
- Mulching (especially in upland rice)
- Herbicides

Common weeds:

- Barnyard grass
- Sedges
- Broadleaf weeds

Pest Management



Pest	Damage	Control
Stem borers	Bore stems causing dead hearts	Resistant varieties, insecticides, field sanitation
Rice bugs	Suck grains	Timely spraying, clean fields
Armyworms	Eat leaves	Biological control, insecticides
Birds	Feed on grains	Bird scaring, nets
Rodents	Destroy plants and grains	Traps, sanitation, baiting
Rice weevils	Damage stored grain	Proper drying and storage

Diseases

Disease	Symptoms	Control
Rice blast	Diamond-shaped lesions on leaves	Resistant varieties, fungicides
Bacterial leaf blight	Yellowing and drying of leaves	Clean seed, resistant varieties
Brown spot	Brown circular spots on leaves	Seed treatment, balanced fertilization
Sheath blight	Lesions on leaf sheaths	Crop rotation, fungicides
False smut	Greenish spore balls on grains	Certified seed, fungicides

Harvesting

Harvest when:

- 80–90% of grains are golden yellow.
- Grain moisture is about 20–24%.

Methods:

- Manual harvesting using sickles.
- Mechanical harvesting using combine harvesters.

Post-Harvest Handling

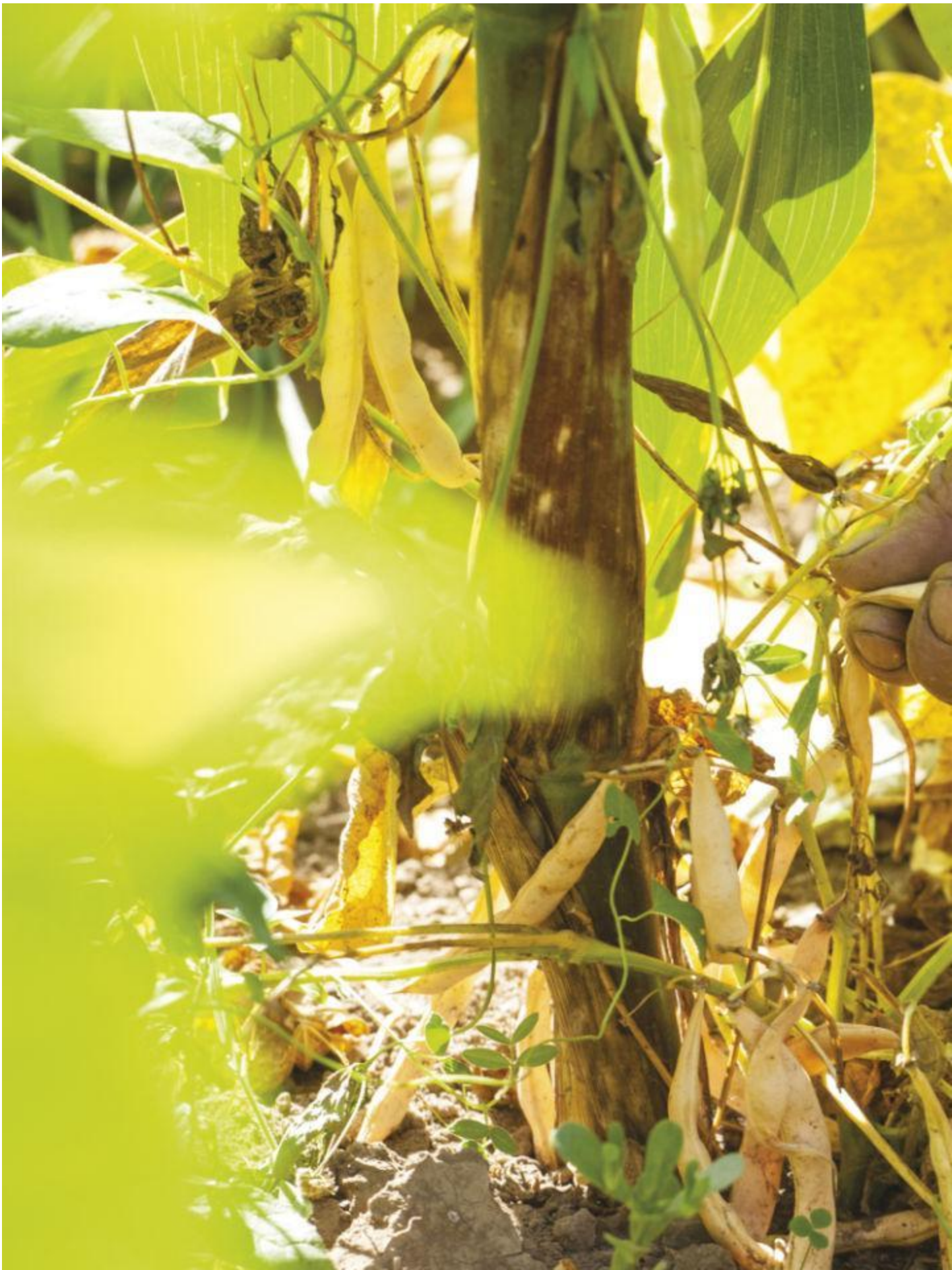
Steps include:

1. Threshing
2. Cleaning
3. Drying (to about 12–14% moisture)
4. Milling
5. Grading
6. Packaging
7. Storage

Proper storage reduces losses from insects, rodents, and mould.

Bean Production

1. Introduction



Beans (*Phaseolus vulgaris*) are among the world's most important legume crops. They are widely grown for their edible seeds and green pods. Beans are rich in protein, carbohydrates, vitamins, minerals, and dietary fiber, making them an important food crop. They also improve soil fertility by fixing atmospheric nitrogen through symbiosis with *Rhizobium* bacteria.

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2. Importance of Beans

Beans are important because they:

- Provide a rich source of protein.
- Supply carbohydrates, vitamins, and minerals.
- Improve soil fertility through nitrogen fixation.
- Generate income for farmers.
- Enhance food and nutritional security.

- Are used as livestock feed (haulms and residues).
- Fit well in crop rotation and intercropping systems.

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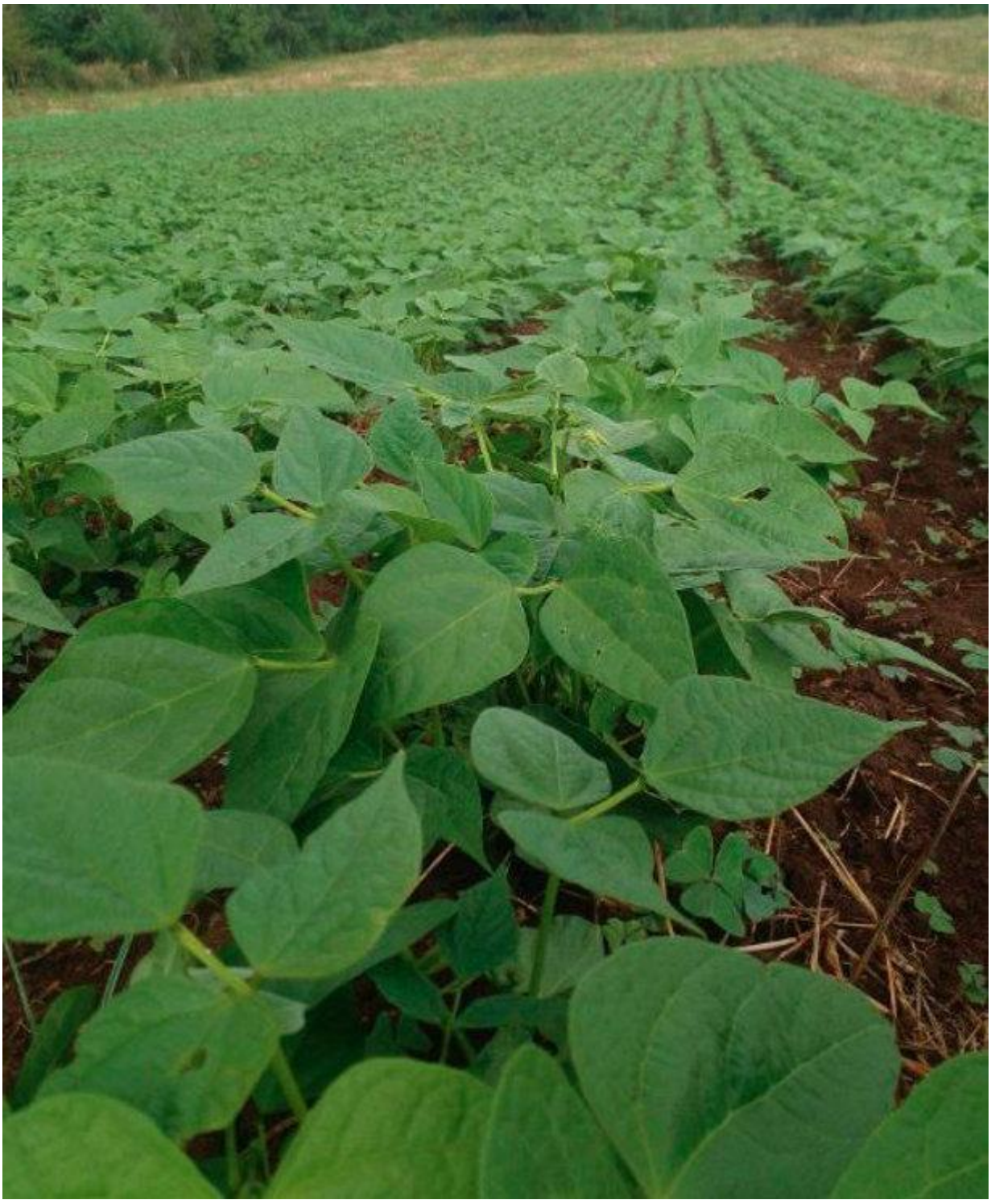
3. Areas Where Beans are Produced

Worldwide

Major bean-producing countries include:

- India
- Brazil
- China
- Mexico
- Tanzania
- Kenya
- Uganda
- Rwanda
- Ethiopia
- Democratic Republic of Congo

In Kenya



Major bean-producing areas include:

- Meru County
- Embu County
- Kirinyaga County
- Nyeri County
- Nyandarua County
- Nakuru County
- Trans Nzoia County
- Uasin Gishu County
- Bungoma County
- Bomet County
- Narok County

- Kisii and Nyamira Counties
- Parts of Machakos and Makueni (with suitable rainfall)

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4. Climatic Requirements

Factor	Requirement
Temperature	18–27°C (Optimum: 20–25°C)
Rainfall	500–1,200 mm, well distributed
Altitude	600–2,100 m above sea level
Sunshine	6–8 hours daily
Humidity	Moderate
Water	Adequate moisture without waterlogging

Beans do not tolerate prolonged flooding or severe drought.

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5. Soil Requirements

Beans grow best in:

- Well-drained loam or sandy loam soils
- Fertile soils rich in organic matter
- Soil pH of **6.0–7.5**
- Deep soils with good aeration

Avoid:

- Waterlogged soils
- Highly acidic soils unless limed

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6. Common Bean Varieties

Examples include:

- Rosecoco

- Canadian Wonder
- Wairimu
- Mwezi Moja
- GLP 2
- Mwitemania
- Yellow Bean
- Chelalang
- KAT B1 (drought tolerant)

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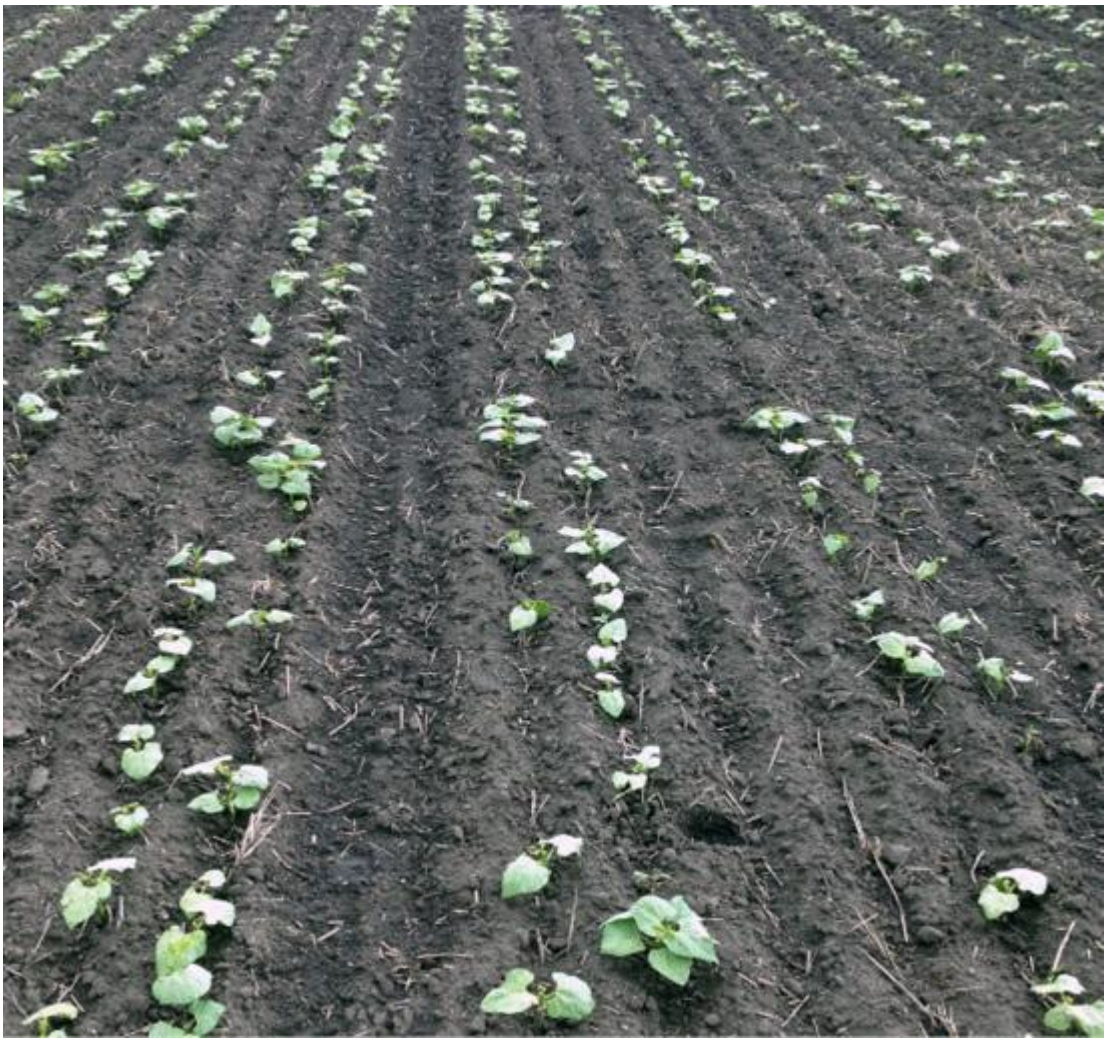
7. Land Preparation



Steps:

1. Clear vegetation.
2. Plough the land.
3. Harrow to obtain a fine seedbed.
4. Remove weeds and crop residues.
5. Mark planting rows.

8. Seed Selection

Good seed should be:

- Certified
- High germination percentage
- Disease-free
- Pure variety
- Free from insect damage

Seed rate:

- **40–80 kg/ha** (depending on variety and seed size)

9. Planting

Beans are planted directly into the field.

Recommended spacing:

- **45–50 cm** between rows
- **10–15 cm** between plants
- Plant **2 seeds per hole**
- Depth: **3–5 cm**

Plant at the onset of rains.

10. Fertilizer and Manure

Recommended inputs:

- Well-decomposed farmyard manure before planting.
- DAP or NPK at planting (based on soil test recommendations).
- Rhizobium inoculant where appropriate to improve nitrogen fixation.
- Lime if soils are acidic.

Beans generally require less nitrogen fertilizer because they fix nitrogen.

11. Field Management Practices

Important practices include:

- Gap filling after germination.
- Early weed control.
- Top dressing only if recommended.
- Earthing up where necessary.
- Mulching to conserve moisture.
- Crop rotation.
- Intercropping with maize.
- Regular scouting for pests and diseases.

Weed Control

Methods include:

- Hand weeding.
- Hoeing.
- Mulching.
- Herbicides where recommended.
- Timely cultivation.

Keep fields weed-free during the first 6–8 weeks after planting.

13. Major Pests

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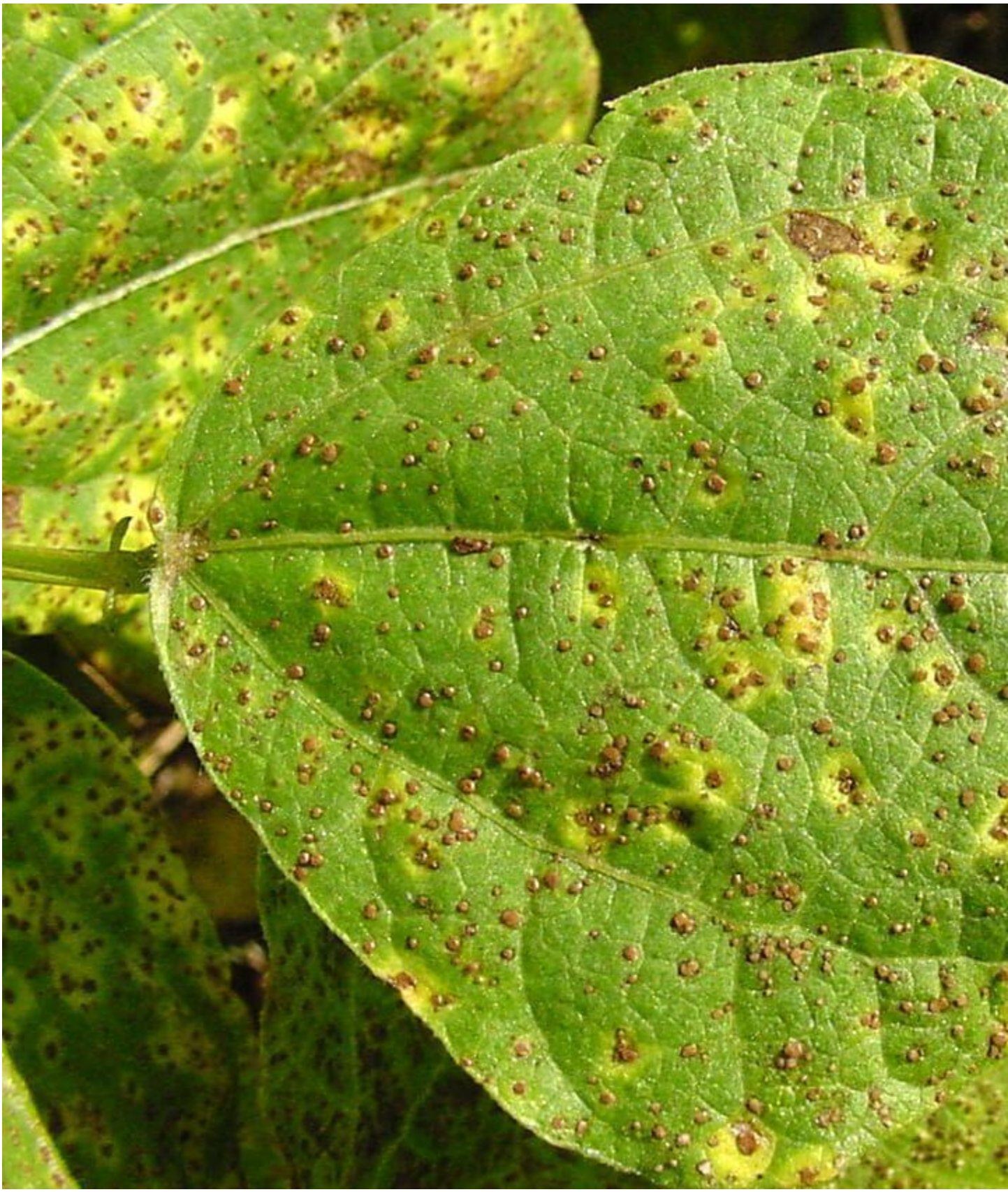
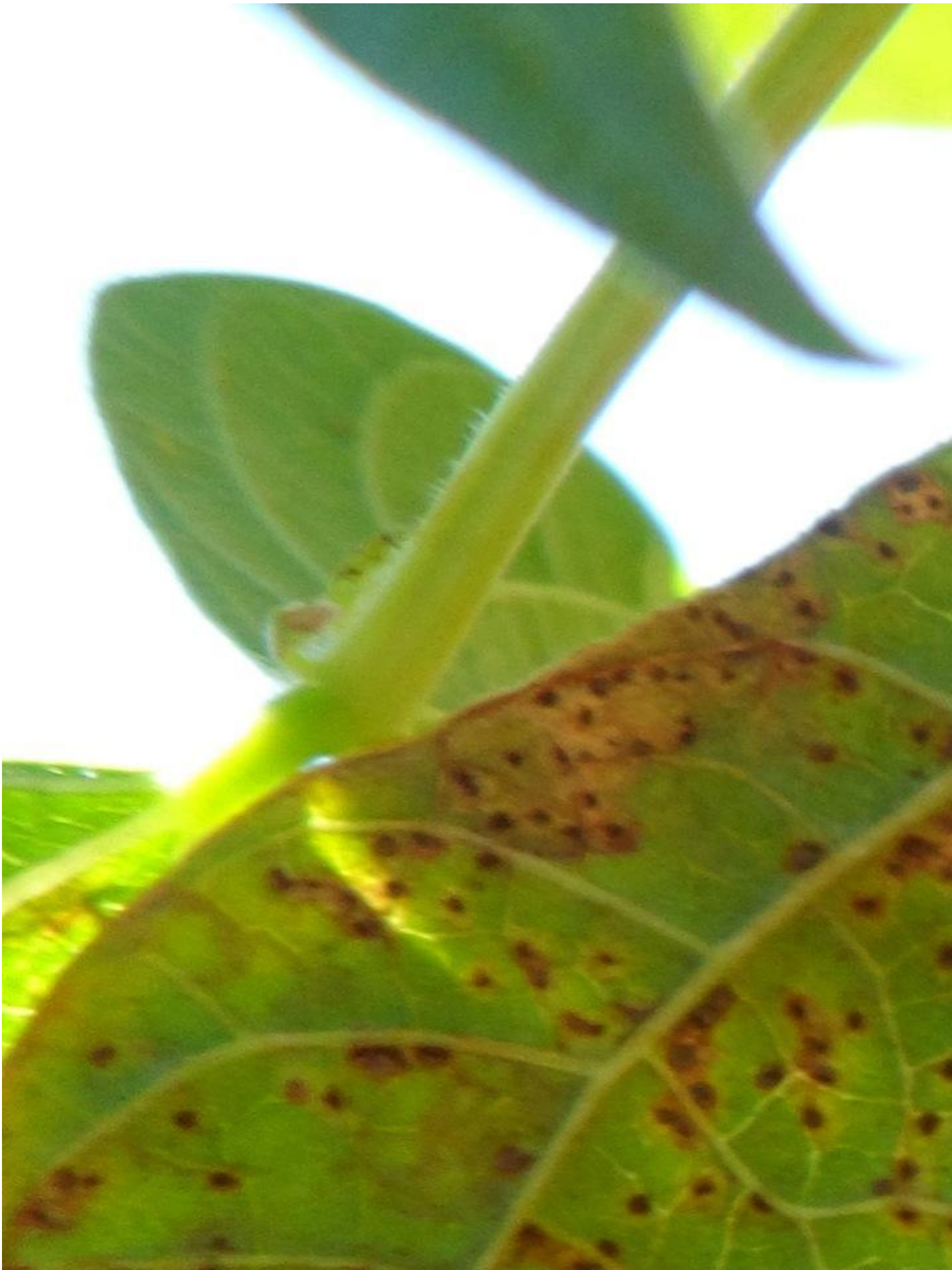
Pest	Damage	Control
Aphids	Suck sap, transmit viruses	Natural enemies, insecticides if needed
Bean stem maggot	Damages stems and seedlings	Early planting, insecticide seed treatment
Bean pod borers	Feed on flowers and pods	Field sanitation, insecticides when necessary
Cutworms	Cut young seedlings	Field hygiene, baiting
Whiteflies	Spread viral diseases	Resistant varieties, insecticides if necessary
Bean weevils	Damage stored grain	Proper drying, airtight storage

14. Major Diseases





Disease	Symptoms	Control
Anthracnose	Dark sunken lesions on stems and pods	Certified seed, resistant varieties
Angular leaf spot	Brown angular lesions on leaves	Crop rotation, fungicides where necessary
Rust	Orange-brown pustules on leaves	Resistant varieties, fungicides
Root rot	Wilting and rotting roots	Good drainage, crop rotation
Bean common mosaic virus	Mottled leaves and stunted growth	Resistant varieties, aphid control

15. Harvesting

Harvest when:

- Pods turn brown and dry.
- Leaves have dried.
- Seeds are hard.

Methods:

- Manual harvesting.
- Mechanical harvesting on large farms.

16. Post-Harvest Handling

Steps:

1. Dry harvested plants or pods.
2. Thresh to separate seeds.
3. Clean the beans.
4. Dry seeds to about **12–13% moisture**.
5. Grade according to size and quality.
6. Store in clean, dry, insect-free containers or hermetic bags.

17. Uses of Beans

Beans are used for:

- Human food (boiled beans, stews, mixed dishes)
- Green beans (vegetable)
- Bean flour
- Livestock feed (haulms and crop residues)
- Improving soil fertility through nitrogen fixation

18. Challenges in Bean Production

- Drought and irregular rainfall
- Poor-quality seed
- Low soil fertility
- Weed competition
- Pest infestations
- Disease outbreaks
- Poor storage leading to bean weevil damage
- High production costs
- Limited access to markets

19. Measures to Improve Bean Production

- Plant certified, high-yielding varieties.
- Prepare land properly.
- Plant at the onset of rains.
- Use recommended spacing.
- Apply manure and fertilizers based on soil tests.
- Practice crop rotation and intercropping.
- Control weeds early.
- Use integrated pest and disease management (IPM).
- Harvest at the correct stage.
- Dry and store beans properly in airtight containers or hermetic bags.

20. Expected Yield

Average yields:

- Traditional farming: **0.5–1.5 tonnes/ha**
- Improved varieties with good management: **2–3 tonnes/ha**
- Under excellent management and favorable conditions: **3–4 tonnes/ha**

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21. Summary

Beans are an important food and cash crop that thrive in warm climates with moderate rainfall and well-drained fertile soils. They are valued for their high protein content and their ability to enrich soil through nitrogen fixation. High yields are achieved through the use of certified seed, timely planting, proper spacing, good soil fertility management, effective weed, pest, and disease control, and careful harvesting and storage.

Irish Potato Production

1. Introduction

Irish potato (*Solanum tuberosum*) is one of the world's most important food crops after maize, rice, and wheat. It is grown for its underground tubers, which are rich in carbohydrates, vitamins (especially vitamin C), minerals, and dietary fiber. Irish potatoes are an important food and cash crop in many highland regions of Kenya.

2. Importance of Irish Potatoes

Irish potatoes are important because they:

- Provide a major source of carbohydrates.
- Supply vitamins, minerals, and dietary fiber.

- Generate income for farmers.
- Create employment in farming, transport, processing, and marketing.
- Contribute to food security.
- Are used in making chips, crisps, mashed potatoes, flour, and starch.
- Provide livestock feed from potato vines and rejected tubers.

3. Areas Where Irish Potatoes are Produced

Worldwide

Major potato-producing countries include:

- China
- India
- Russia
- Ukraine
- United States
- Germany
- Poland
- Bangladesh
- Netherlands
- Peru

In Kenya

Major potato-growing areas include:

- Nyandarua County (leading producer)
- Nakuru County
- Meru County
- Nyeri County
- Elgeyo Marakwet County
- Bomet County
- Narok County
- Kiambu County
- Trans Nzoia County
- Uasin Gishu County
- Kericho County

4. Climatic Requirements

Factor	Requirement
Temperature	15–20°C (Optimum: 18°C)
Rainfall	750–1,200 mm, well distributed
Altitude	1,500–3,000 m above sea level
Sunshine	6–8 hours daily
Humidity	Moderate
Water	Adequate moisture without waterlogging

High temperatures reduce tuber formation and yield.

5. Soil Requirements

Irish potatoes grow best in:

- Deep, fertile loam or sandy loam soils
- Well-drained soils
- Soils rich in organic matter
- Soil pH of **5.0–6.5**
- Loose soils that allow tubers to expand

Avoid:

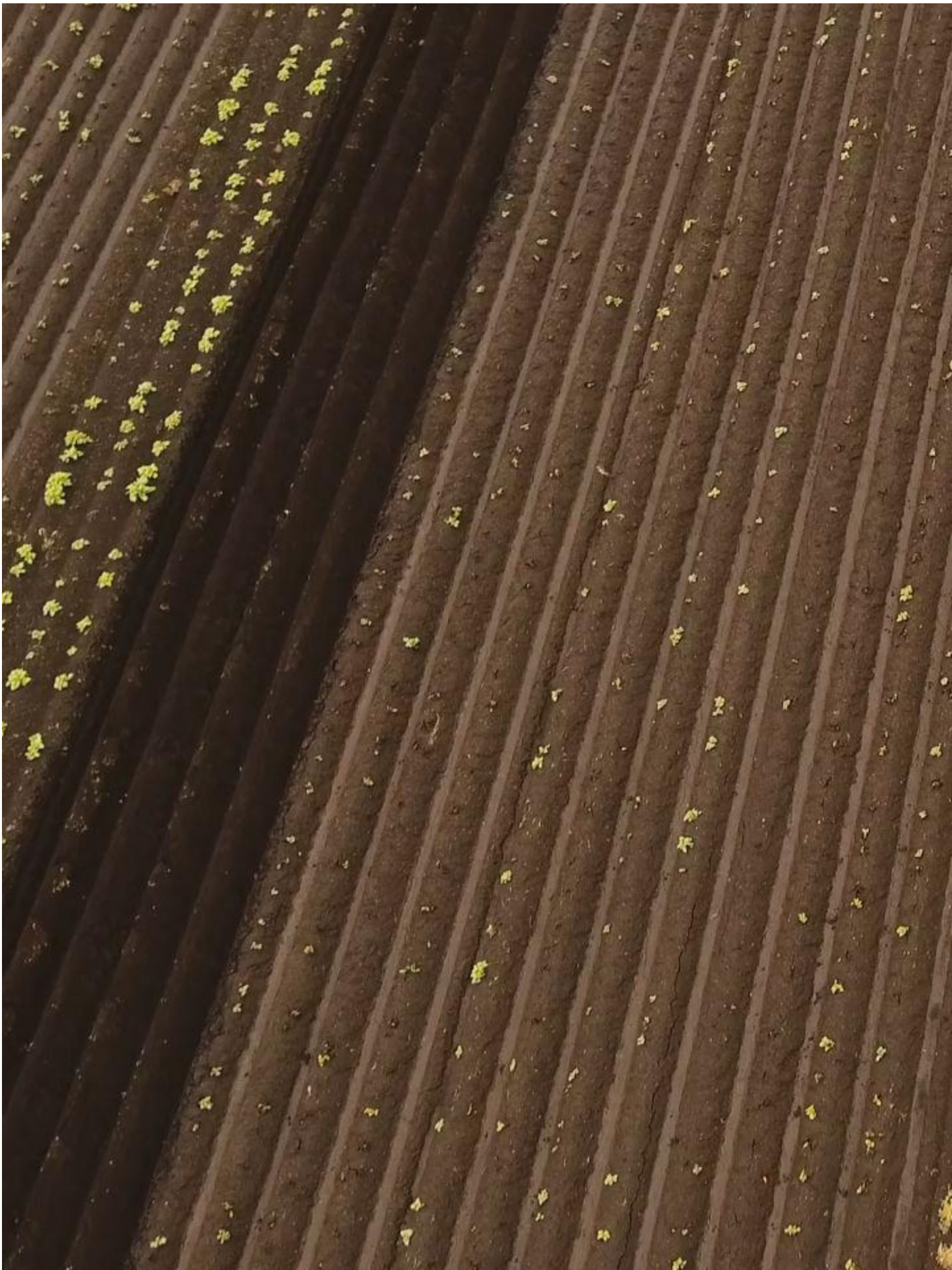
- Heavy clay soils with poor drainage
- Waterlogged fields

6. Common Varieties

Examples include:

- Shangi
- Unica
- Tigoni
- Asante
- Kenya Mpya
- Dutch Robijn
- Desiree
- Markies
- Dutch Robjin
- Wanjiku

7. Land Preparation



Steps:

1. Clear vegetation.
2. Plough the land deeply.
3. Harrow to produce a fine tilth.
4. Incorporate well-decomposed manure.
5. Make ridges approximately **75 cm apart**.

8. Seed Selection

Use:

- Certified seed tubers.
- Healthy disease-free tubers.
- Medium-sized tubers (30–60 g).
- Tubers with well-developed sprouts.

Avoid:

- Cut or diseased seed tubers.
- Tubers affected by bacterial wilt or late blight.

Seed rate:

- **2–3 tonnes of seed tubers per hectare**

9. Planting

Recommended spacing:

- **75 cm** between rows
- **30 cm** between plants
- Planting depth: **10–15 cm**

Plant at the beginning of the rainy season or under irrigation.

10. Fertilizer and Manure

Apply:

- Well-decomposed farmyard manure before planting.
- DAP or NPK fertilizer at planting (based on soil test recommendations).
- Top-dress with CAN where needed after crop establishment.

Important nutrients:

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)

11. Field Management Practices

Essential practices include:

- Gap filling.
- Regular weeding.
- Earthing up (hilling) to cover developing tubers and reduce greening.
- Irrigation during dry periods.
- Mulching where appropriate.
- Timely fertilizer application.
- Crop rotation.
- Regular pest and disease scouting.

12. Weed Control

Methods include:

- Hand weeding.
- Hoeing.
- Mulching.
- Herbicides where recommended.
- Early weed removal to reduce competition.

13. Major Pests

Pest	Damage	Control
Potato tuber moth	Damages tubers and leaves	Earthing up, field sanitation, insecticides where needed
Aphids	Suck sap and transmit viruses	Natural enemies, resistant varieties, insecticides if necessary
Cutworms	Cut young plants	Baiting, field sanitation
White grubs	Feed on roots and tubers	Crop rotation, soil management
Leaf miners	Damage leaves	Monitoring and appropriate insecticides if necessary

14. Major Diseases

HUGOT N. (C.A. NOUVELLE-CALÉDONIE)

Disease	Symptoms	Control
Late blight	Dark lesions on leaves and stems, tuber rot	Certified seed, fungicides, resistant varieties
Early blight	Brown spots with concentric rings	Crop rotation, fungicides
Bacterial wilt	Sudden wilting, brown vascular tissue	Disease-free seed, crop rotation, sanitation
Potato leaf roll virus	Upward rolling leaves, stunted growth	Control aphids, certified seed
Black scurf	Black crusts on tubers	Certified seed, crop rotation

15. Harvesting

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Harvest when:

- Leaves turn yellow and dry.
- Tuber skins are firm and do not peel easily.

Methods:

- Manual harvesting using hoes or forks.
- Mechanical harvesting on large farms.

Typical maturity:

- **90–150 days** after planting, depending on the variety.

16. Post-Harvest Handling

Steps:

1. Remove soil from tubers.
2. Sort and grade potatoes.
3. Cure tubers in a cool, shaded, well-ventilated place for about 10–14 days to heal minor wounds.
4. Store in a cool, dark, dry, and well-ventilated store.
5. Remove damaged or diseased tubers to reduce spoilage.

17. Uses of Irish Potatoes

Irish potatoes are used for:

- Boiling
- Chips (French fries)
- Crisps
- Mashed potatoes
- Potato flour
- Starch production
- Livestock feed (damaged tubers and vines)

18. Challenges in Irish Potato Production

- Late blight outbreaks
- Bacterial wilt
- High cost of certified seed
- Low soil fertility
- Drought and erratic rainfall
- Frost in some highland areas
- Poor storage facilities
- Pest infestations
- Market price fluctuations

19. Measures to Improve Production

- Use certified disease-free seed tubers.
- Practice crop rotation with non-solanaceous crops.
- Plant resistant or tolerant varieties.
- Apply manure and fertilizers according to soil test recommendations.
- Carry out timely earthing up.
- Control weeds, pests, and diseases early using integrated pest management (IPM).
- Harvest at the correct maturity stage.
- Store potatoes under proper conditions to reduce post-harvest losses.

20. Expected Yield

Average yields:

- Traditional farming: **8–15 tonnes/ha**
- Improved varieties with good management: **20–35 tonnes/ha**
- Well-managed commercial production: **35–50 tonnes/ha or more**

21. Summary

Irish potatoes are a valuable food and cash crop that perform best in cool, high-altitude areas with fertile, well-drained soils and reliable rainfall. Successful production depends on using certified seed tubers, proper land preparation, correct spacing, balanced nutrient management, timely earthing up, and effective control of weeds, pests, and diseases. Good harvesting and storage practices help maintain tuber quality and minimize post-harvest losses, resulting in higher yields and better income for farmers.

Coffee Production

1. Introduction

Coffee (*Coffea* spp.) is one of the world's most valuable cash crops and a major source of income for millions of farmers. It is grown mainly for its seeds.

Coffee Production (Continued)

2. Importance of Coffee

Coffee is important because it:

- Generates foreign exchange through exports.
- Provides income to farmers.
- Creates employment in farming, processing, transportation, and marketing.
- Contributes to national economic growth.
- Supports many agro-industries such as roasting, grinding, and beverage production.
- Coffee husks and pulp can be used for compost, organic manure, and biofuel production.

3. Types of Coffee

There are two major commercial species:

A. Arabica Coffee (*Coffea arabica*)

- Produces high-quality coffee with excellent flavor and aroma.
- Grown in cool, high-altitude areas.
- Accounts for about 60–70% of world coffee production.

B. Robusta Coffee (*Coffea canephora*)

- Higher yielding than Arabica.
- More tolerant to pests, diseases, and higher temperatures.
- Contains more caffeine.
- Commonly used in instant coffee production.

4. Areas Where Coffee is Produced

Worldwide

Major coffee-producing countries include:

- Brazil
- Vietnam
- Colombia
- Ethiopia
- Indonesia
- Uganda
- Honduras
- India
- Peru
- Guatemala

In Kenya

Major coffee-growing counties include:

- Kiambu
- Murang'a
- Nyeri
- Kirinyaga
- Embu
- Meru
- Nyandarua
- Nakuru
- Machakos
- Bungoma
- Kisii
-

5. Climatic Requirements

Factor	Requirement
Temperature	15–24°C (Arabica), 24–30°C (Robusta)
Rainfall	1,200–2,000 mm per year, well distributed

Factor	Requirement
Altitude	Arabica: 1,200–2,100 m; Robusta: 0–900 m
Relative Humidity	70–90%
Sunshine	Moderate sunshine with partial shade in some areas
Wind	Moderate; strong winds should be controlled using windbreaks

6. Soil Requirements

Coffee grows best in:

- Deep, fertile, well-drained volcanic soils.
- Loam or clay loam soils.
- Soil rich in organic matter.
- Soil pH of **5.5–6.5**.
- Good water-holding capacity.

Avoid:

- Waterlogged soils.
- Saline soils.
- Very shallow soils.

7. Common Coffee Varieties

Arabica

- SL28
- SL34
- Ruiru 11
- Batian
- K7

Robusta

- Robusta clones
- Elite Robusta varieties

8. Land Preparation

Steps include:

1. Clear bushes and weeds.
2. Plough if necessary.
3. Dig planting holes (about **60 cm × 60 cm × 60 cm**).
4. Mix topsoil with well-decomposed manure.
5. Plant at the onset of the rainy season.

9. Planting

Recommended spacing:

Variety	Spacing
Arabica	2 m × 2 m or 2.5 m × 2.5 m
Robusta	3 m × 3 m

Plant healthy certified seedlings from reputable nurseries.

10. Fertilizer and Manure

Apply:

- Well-decomposed farmyard manure.
- Compound NPK fertilizers according to soil test recommendations.
- Lime where soils are acidic.

Important nutrients:

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Calcium (Ca)
- Magnesium (Mg)

11. Field Management Practices

Important practices include:

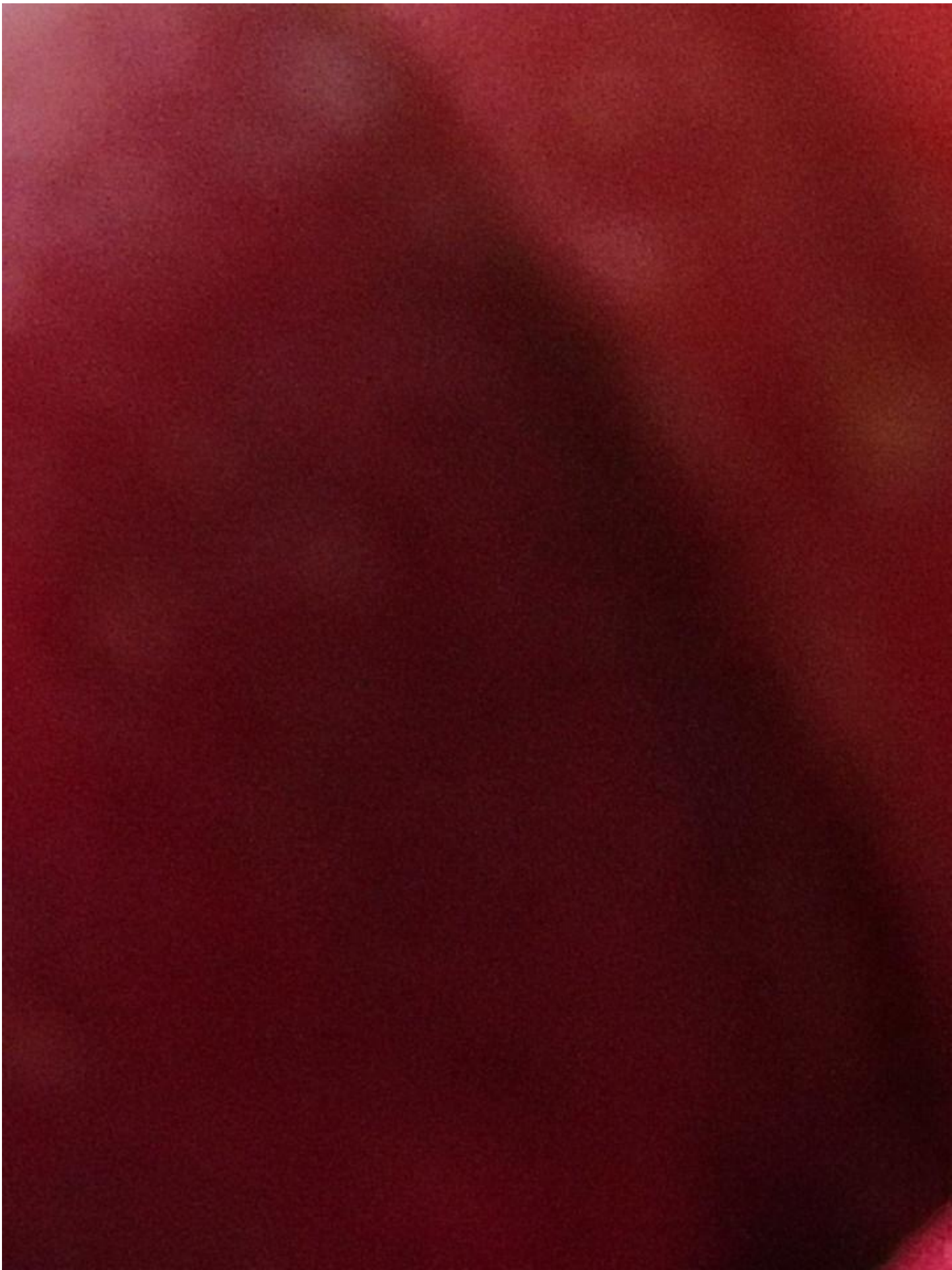
- Regular pruning.
- Mulching.
- Weeding.
- Fertilizer application.
- Irrigation during prolonged dry periods.
- Shade management where appropriate.
- Removal of diseased branches.
- Soil conservation on sloping land.
- Regular scouting for pests and diseases.

12. Weed Control

Methods include:

- Hand weeding.
- Slashing.
- Mulching.
- Cover cropping.
- Herbicides where recommended.

Major Pests

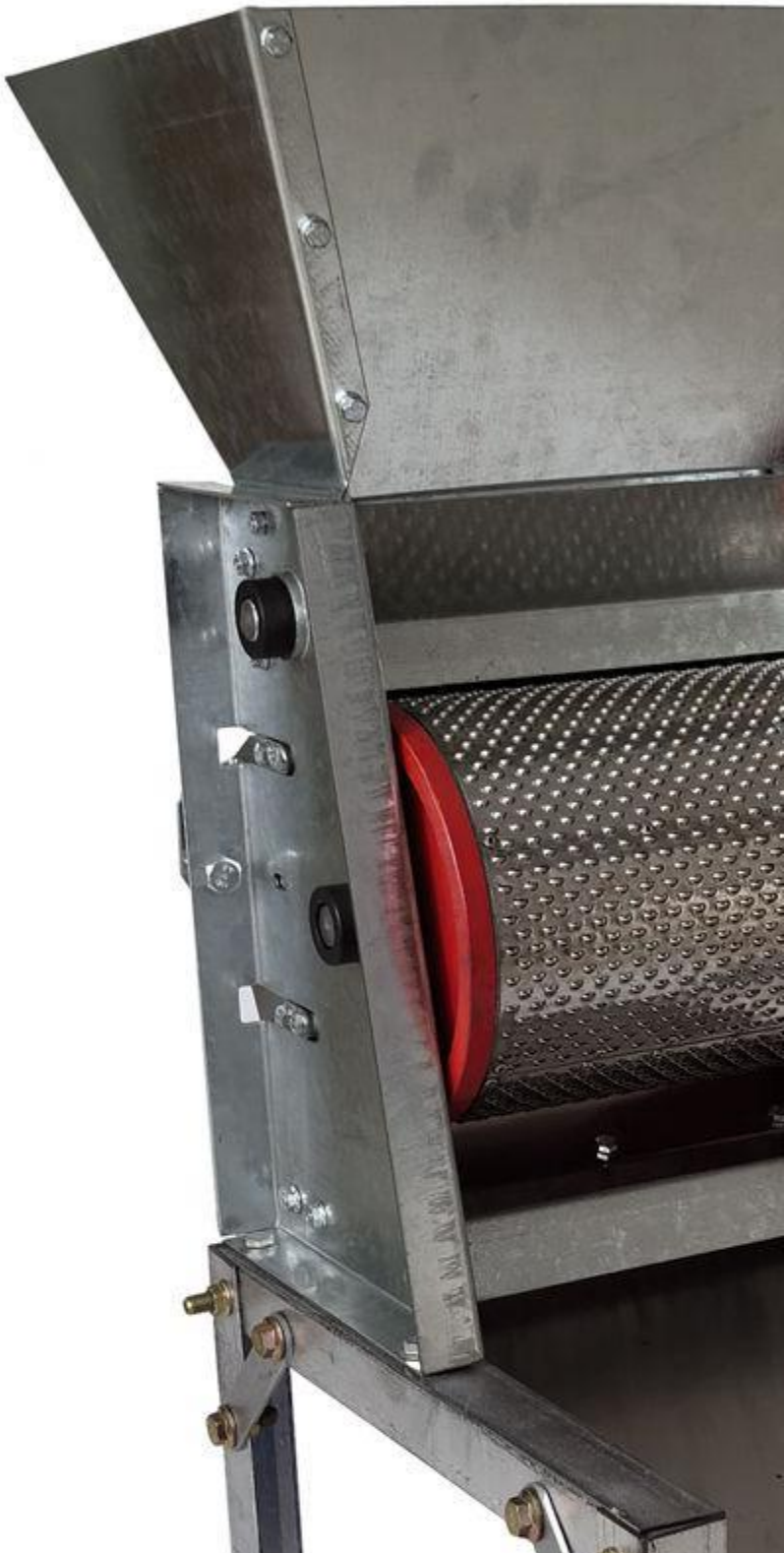


Pest	Damage	Control
Coffee berry borer	Bores into berries, reducing quality and yield	Field sanitation, timely harvesting, approved insecticides
Stem borers	Tunnel into stems causing wilting and death	Remove affected plants, insecticides, field hygiene
Leaf miners	Feed within leaves, reducing photosynthesis	Monitoring and insecticides where necessary
Green scales	Suck plant sap	Encourage natural enemies, prune infested branches, approved insecticides
Mealybugs	Weaken plants by sucking sap	Field sanitation and biological or chemical control where needed

Major Diseases

Disease	Symptoms	Control
Coffee leaf rust	Orange powdery spots on leaves, premature leaf fall	Resistant varieties, fungicides, good nutrition
Coffee berry disease	Black sunken lesions on berries	Resistant varieties, timely fungicide application
Coffee wilt disease	Wilting and death of trees	Remove infected plants, sanitation
Bacterial blight	Dark lesions on leaves and shoots	Pruning, sanitation, resistant varieties where available

Harvesting



Harvest when:

- Cherries are fully ripe (bright red or yellow depending on the variety).
- Harvesting is usually done by hand to ensure only ripe cherries are picked.

Post-Harvest Handling

Steps include:

1. Sorting ripe cherries.
2. Pulping.
3. Fermentation (for washed coffee).
4. Washing.
5. Drying beans to about **10–12% moisture**.
6. Hulling.
7. Grading.
8. Packaging.
9. Storage in cool, dry conditions.

Proper post-harvest handling preserves bean quality and market value.

Uses of Coffee

Coffee is used for:

- Preparing beverages.
- Production of instant coffee.
- Coffee-flavored foods and desserts.
- Cosmetics and skincare products (using coffee extracts).
- Organic manure and compost from coffee pulp.
- Biofuel production from coffee waste.

Challenges in Coffee Production

- Coffee berry disease.
- Coffee leaf rust.
- Coffee berry borer infestation.
- Climate change and prolonged drought.
- Fluctuating world coffee prices.
- High labour costs.
- Poor soil fertility.
- Aging coffee trees.
- Limited access to quality seedlings and farm inputs.

Measures to Improve Coffee Production

- Plant certified, high-yielding, disease-resistant varieties such as Batian and Ruiru 11.
- Apply fertilizers based on soil test results.
- Mulch to conserve soil moisture.
- Prune regularly to improve growth and yield.
- Practice integrated pest and disease management (IPM).
- Harvest only ripe cherries.
- Process and dry beans properly to maintain quality.
- Replace old, unproductive trees with improved varieties.
- Adopt water conservation and irrigation where feasible.

Expected Yield

Average yields:

- Traditional farms: **500–1,000 kg of clean coffee per hectare**
- Well-managed farms: **1,500–3,000 kg per hectare**
- Intensive commercial production with improved varieties: **More than 3,000 kg per hectare**

Summary

Coffee is a high-value perennial cash crop that grows best in fertile, well-drained soils under warm conditions with adequate rainfall. Successful coffee production depends on selecting suitable varieties, proper land preparation, good field management, balanced fertilization, effective pest and disease control, and careful harvesting and post-harvest processing. With good agronomic practices, coffee can provide high-quality beans, sustainable incomes for farmers, and significant foreign exchange earnings for producing countries.